

H, *W*, 3

Stage 0 — ConvHead, 16

H/2, *W*/2, 16

Stage 1 — Max-Pool/2; DWBlock, 16; DWBlock, 64

H/4, *W*/4, 64

Stage 2 — DWBlock, 64; Max-pool/2

H/8, *W*/8, 64

Stage 3 — DWBlock, 64; Max-pool/2

H/16, *W*/16, 64

Stage 4 — DWBlock, 64; Max-pool/2

H/32, *W*/32, 64

Backbone

DWUnit = Conv1×1, *g* = 1; Conv3×3, *p* = 1, *g* = out; BN & Relu

DWBlock = DWUnit; DWUnit

ConvHead = Conv3×3, *s* = 2, *p* = 1; BN & Relu; DWUnit

DecoupledHead = DWUnit-cls, 1; DWUnit-obj, 1; DWUnit-bbox, 4; DWUnit-kps, 10

Upsample

Upsample

DWUnit, 64 → DWUnit, 64 → DecoupledHead

H/8, *W*/8, 64

DWUnit, 64 → DWUnit, 64 → DecoupledHead

H/16, *W*/16, 64

DWUnit, 64 → DWUnit, 64 → DecoupledHead

H/32, *W*/32, 64

TFPN

Head

Post-process